

HDOC Day 21

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Batch-15

1. Algorithm

1. Start.
2. Declare integer variables **m**, **n**, **choice**, **i**, and **j**.
3. Input **m** and **n**.
4. Validate the input:
 - Both **m** and **n** must be positive.
 - **m** <= **n**.
 - If invalid, display an error message and ask again.
5. Display the menu:
 - 1 – First 10 multiples
 - 2 – Addition tables
 - 3 – Subtraction tables
 - 4 – Division tables
 - 5 – Exit
6. Input **choice**.
7. Use **switch(choice)**:
 - **Case 1:** For every integer from **m** to **n**, use an inner loop from 1 to 10 and print **i * j**.
 - **Case 2:** For every integer from **m** to **n**, use an inner loop from 1 to 10 and print **i + j**.
 - **Case 3:** For every integer from **m** to **n**, use an inner loop from 1 to 10 and print **i - j**.
 - **Case 4:** For every integer from **m** to **n**, use an inner loop from 1 to 10 and print **(double) i / j** rounded to two decimal places.
 - **Case 5:** Display the exit message.
 - **Default:** Display an invalid menu choice error.
8. Repeat Steps 5–7 until the user chooses 5.
9. Stop.

2. Test-Case Table

Test Case	Input	Expected Output	Actual Output	Result
1 – Valid multiples	m=2, n=3, choice=1	Tables of 2 and 3 from $\times 1$ to $\times 10$	Correct tables displayed	Pass
2 – Boundary m=n	m=1, n=1, choice=2	$1+1=2$ through $1+10=11$	Correct addition table	Pass
3 – Valid subtraction	m=5, n=6, choice=3	Subtraction tables of 5 and 6	Correct tables displayed	Pass
4 – Valid division	m=3, n=3, choice=4	$3/1=3.00 \dots 3/10=0.30$	Correct results to 2 decimal places	Pass
5 – Invalid menu choice	m=2, n=2, choice=9	Invalid menu choice message; menu displayed again	Error displayed correctly	Pass
6 – Invalid positive input	m=0, n=4	Error and request input again	Error displayed correctly	Pass
7 – Invalid	m=5, n=3	Error because $m > n$; request	Error displayed	Pass
8 – Exit	m=2, n=2, choice=5	Exiting program.	Program exits normally	Pass

3. C Program

```
#include <stdio.h>
```

```
int main(void)
```

```
{
```

```
    int m, n, choice, i, j;
```

```
    /* Input and validation of m and n */
```

```
    do
```

```
    {
```

```
        printf("Enter two positive integers m and n (m <= n):
```

```
");
```

```
        scanf("%d %d", &m, &n);
```

```
        if (m <= 0 || n <= 0 || m > n)
```

```
            printf("Invalid input! Enter positive integers  
such that m <= n.\n");
```

```
    } while (m <= 0 || n <= 0 || m > n);
```

```
    do
```

```

{
    printf("\nMENU\n");
    printf("1. Print first 10 multiples\n");
    printf("2. Print addition tables\n");
    printf("3. Print subtraction tables\n");
    printf("4. Print division tables\n");
    printf("5. Exit\n");

    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            for (i = m; i <= n; i++)
            {
                printf("\nMultiplication table of %d:\n",
i);

                for (j = 1; j <= 10; j++)
                {
                    printf("%d x %d = %d\n",
                        i, j, i * j);
                }
            }
            break;

        case 2:
            for (i = m; i <= n; i++)
            {
                printf("\nAddition table of %d:\n", i);

                for (j = 1; j <= 10; j++)
                {
                    printf("%d + %d = %d\n",
                        i, j, i + j);
                }
            }
            break;

        case 3:
            for (i = m; i <= n; i++)
            {
                printf("\nSubtraction table of %d:\n",
i);

```

```

        for (j = 1; j <= 10; j++)
        {
            printf("%d - %d = %d\n",
                    i, j, i - j);
        }
    }
    break;

case 4:
    for (i = m; i <= n; i++)
    {
        printf("\nDivision table of %d:\n", i);

        for (j = 1; j <= 10; j++)
        {
            printf("%d / %d = %.2f\n",
                    i, j, (double)i / j);
        }
    }
    break;

case 5:
    printf("Exiting program.\n");
    break;

default:
    printf("Invalid menu choice! "
           "Please enter a number from 1 to
5.\n");
}

    } while (choice != 5);

    return 0;
}

```

4. Compilation

I compiled the program using GCC with:

```
gcc menu_tables.c -Wall -Wextra -std=c11 -o menu_tables
```

Compilation result: SUCCESS — no compilation errors.

5. Execution

```
[aaryanrajput@aaryans-macbook ~ % gcc menu_tables.c -Wall -Wextra -std=c11 -o menu_tables
[aaryanrajput@aaryans-macbook ~ % ./menu_tables.c
zsh: permission denied: ./menu_tables.c
[aaryanrajput@aaryans-macbook ~ % ./menu_tables
Enter two positive integers m and n (m <= n): 2 3

MENU
1. Print first 10 multiples
2. Print addition tables
3. Print subtraction tables
4. Print division tables
5. Exit
Enter your choice: 1

Multiplication table of 2:
2 x 1 = 2
2 x 2 = 4
2 x 3 = 6
2 x 4 = 8
2 x 5 = 10
2 x 6 = 12
2 x 7 = 14
2 x 8 = 16
2 x 9 = 18
2 x 10 = 20

Multiplication table of 3:
3 x 1 = 3
3 x 2 = 6
3 x 3 = 9
3 x 4 = 12
3 x 5 = 15
3 x 6 = 18
3 x 7 = 21
3 x 8 = 24
3 x 9 = 27
3 x 10 = 30

MENU
1. Print first 10 multiples
2. Print addition tables
3. Print subtraction tables
4. Print division tables
5. Exit
Enter your choice: 5
Exiting program.
aaryanrajput@aaryans-macbook ~ %
```